MULTIPLAYER HUMAN-AGENT COLLABORATION

Ignacio Guarna
Stanford University
iguarna@stanford.edu

ABSTRACT

AI assistants have demonstrated impressive capabilities in supporting individuals on a wide range of text-based tasks, yet there has been relatively limited progress in integrating these assistants into multi-human collaboration settings. From a human-centered design perspective, agents must enhance and foster human collaboration rather than promote isolated work. In this work, we propose a framework to train collaboration facilitator assistants, capable of intervening in contentious comment threads to help drive the discussion to more efficient and constructive resolution. We create a synthetic dataset of simulated collaborative writing stalemates, and utilize preference labels from an LLM-as-a-judge to align a 4B-parameter open-weights model via Direct Preference Optimization (DPO). Our results demonstrate that while DPO significantly improves the agent's win-rate against its base instruction-tuned counterpart, the model struggles to implicitly learn an intervention selection policy solely from preference labels. Furthermore, we provide a quantitative analysis of LLM-as-a-judge limitations and propose new research directions to advance collaboration facilitator assistants.

1 INTRODUCTION

AI assistants have risen to become a fundamental tool in writing tasks. Reasoning capabilities and tool usage are advancing at a rapid pace, making AI agents reliable 'copilots' for a wide range of activities. However, current assistant implementations are largely designed to align with the intent of a single user and follow their instructions.

The single-user nature of current human-AI interaction contrasts with the collaborative essence of most knowledge-based professions. From a human-centered perspective, AI assistants must support and enhance human collaboration. Yet, there has been relatively limited progress in integrating these assistants into multi-human collaboration settings, and the prevailing single-user paradigm often reinforces isolated workflows.

In this work, we explore a framework for developing collaboration facilitator assistants, capable of intervening in contentious comment threads to help drive the discussion to more efficient and constructive resolution. The main challenge in training agents for this task is the scarcity of large-scale collaborative writing datasets, and thus the need to synthesize human-like conflict data.

While frontier Large Language Models (LLMs) like GPT-4.1 exhibit strong reasoning capabilities, their high inference cost and latency make them impractical for real-time collaborative environments where multiple inference runs are needed in a session. Furthermore, sending sensitive drafts to external APIs raises significant privacy concerns. We posit that facilitating collaboration is a specialized skill that does not require the vast knowledge and reasoning capabilities of multi-trillion-parameter models. Therefore, we focus on aligning a relatively small, open-weights model (4B parameters). This approach offers a path toward privacy-preserving, low-latency agents that can run locally on user devices, democratizing access to AI-assisted collaboration.

In this paper, we study the following research questions:

RQ1 Can a small, open-weights model (4B parameters) be aligned to act as a collaboration facilitator using only synthetic data and DPO?

RQ2 Is the preference signal from an LLM-as-a-Judge sufficient to teach a small model how to select the most appropriate intervention type in each context, or does the training only result in generic helpfulness improvements without altering the intervention type distribution?

RQ3 Can proprietary LLMs serve as reliable evaluators for this domain, and what limitations arise when using larger models to judge smaller, specialized assistants?

Contributions. We present the following contributions:

1. **A Synthetic Task Conflict Dataset**. We introduce a methodology for generating realistic, multi-turn collaborative writing stalemates that can be used to model task conflict.
2. **Empirical Validation of DPO**. We demonstrate that DPO significantly improves a 4B model's win-rate against its instruction-tuned baseline, based on the evaluation of an LLM-as-a-judge.
3. **Analysis of Implicit Strategy Learning**. We analyze the distribution of intervention types (questioning, synthesizing, and no intervention) before and after training. We find that DPO failed to alter the distribution of the base model, meaning the agent learned to generate "better" interventions within a type, but did not learn to dynamically switch strategies based on the context. This suggests that preference optimization alone is insufficient for learning high-level conflict mediation policies.
4. **Evaluation of LLM-as-a-judge for Conflict Modeling**. We provide a quantitative analysis of the limitations of using LLM-as-a-judge for social alignment, identifying consistency issues, and proposing future architectural improvements.

2 Related work

This project builds on and extends several lines of research in human-AI collaboration and addresses a critical gap: agents that can proactively support multi-human collaboration rather than simply responding to individual users' requests.

Human-agent collaboration in writing. Prior work has explored various aspects of AI-assisted writing, predominantly focusing on single-user alignment. Ge Gao (2025) developed a framework for inferring user preferences from edits, but their approach remains fundamentally single-user and does not address the complexities of multiple humans collaborating with different objectives. More recently, Yijia Shao (2025) introduced Collaborative Gym (Co-Gym), moving beyond rigid turn-taking to allow simultaneous human-agent interaction in shared workspaces. While Co-Gym enables more fluid collaboration, it optimizes for task efficiency in a dyadic setting (one human, one agent). Our work diverges by positioning the agent not as a co-author, but as a mediator between multiple human authors, optimizing for consensus rather than accuracy or speed.

Multi-user agent systems. The limitations of single-user agents have been identified in prior works. Florian Lehmann (2025) proposes a comment-based interface where multiple human collaborators can invoke agents by "mentioning" them in discussion threads. However, their framework positions the agent as a reactive assistant that responds only when explicitly called upon to perform tasks like grammar checking or drafting. In contrast, our work explores proactive agent interventions. Rather than waiting for a prompt, our proposed agent monitors the social state of the collaboration to identify stalemates, contributing synthesis or clarifying questions without explicit invocation.

AI to support human collaboration. The potential for AI to enhance teamwork dynamics has been demonstrated in other domains outside of text editing. Sangwon Seo (2025) introduced an AI coach that monitors team behavior in real-time and intervenes when it detects misalignment in shared mental models. They validate Socratic questioning as an effective intervention strategy, though in gamified environments far removed from document co-editing. In the domain of civic discourse, Hope Schroeder (2024) presented Fora, a framework for analyzing facilitated dialogue that taxonomizes key human behaviors such as "making connections" and offering "invitations to participate". While Fora provides a granular analysis of general facilitation dynamics, our work operationalizes these concepts into specific intervention strategies—specifically compromise synthesis and disagreement summarization—that are specific to the domain of collaborative writing stalemates.

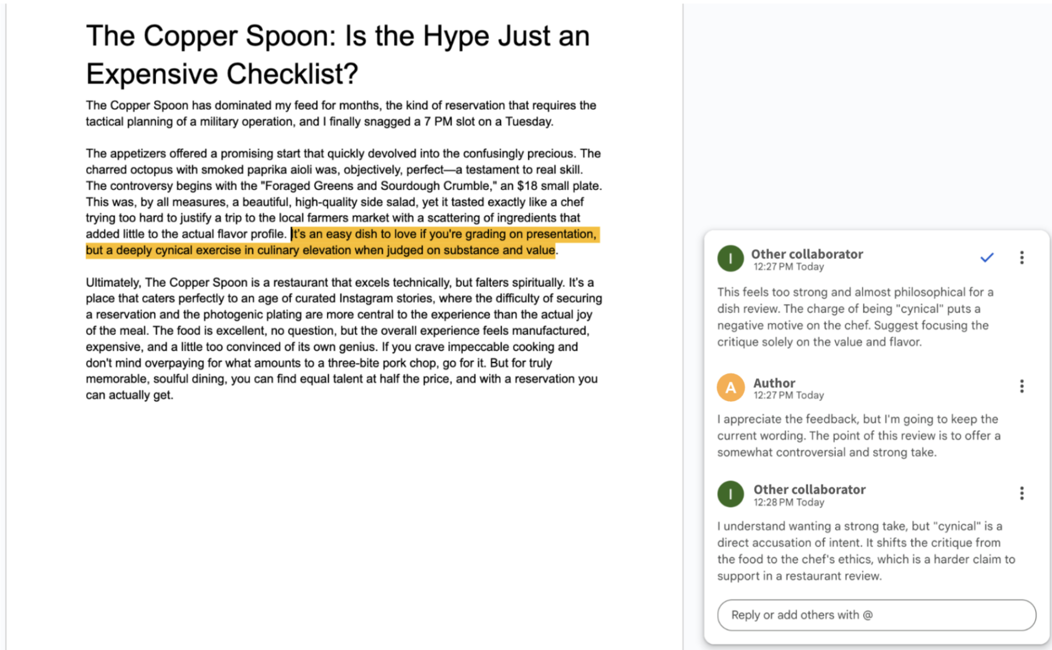

Figure 1: Context artifacts generated by the task conflict simulator.

3 DATA

Because professional collaborative writing often occurs in confidential settings, high-quality datasets of collaborative conflicts are private and scarce. To address this gap, we relied on a synthetic data generation pipeline to create a diverse corpus of realistic writing stalemates.

We designed a simulated environment consisting of short documents of up to three paragraphs across varying domains. With each of the documents, there is a multi-turn comment thread between two simulated users that is focused on a specific sentence in the text. Figure 1 shows an example of the collaborative environment described by the dataset.

The data generation pipeline (illustrated in Figure 1) consists of three stages:

1. **Context generation**: The Task Conflict Simulator generates a document and a comment thread in which two fictional users discuss a specific sentence.
2. **Candidate generation**: This context, along with a prompt, is presented to two AI assistants to generate a comment to intervene in the discussion (detailed in Methods).
3. **Preference labeling**: a "judge" LLM is presented with the two comments along with the context and is instructed to label which is accepted as the most effective intervention, resulting in a triplet dataset of $(x, y_{accepted}, y_{rejected})$ for DPO training.

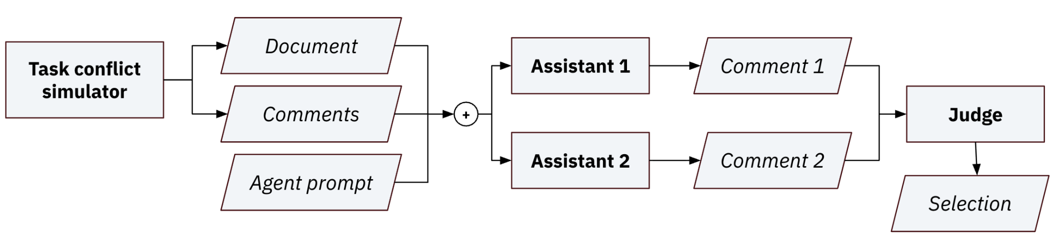

Figure 2: The data generation pipeline for conflict task modeling.

3.1 Task conflict simulator

The simulator generates the contextual conflict and the resulting discussion stalemate through the iterative process illustrated in Figure 3. We utilized *gpt-4.1-mini* (OpenAI) for this stage, as pilot experiments with larger models showed no significant quality difference for this specific sub-task.

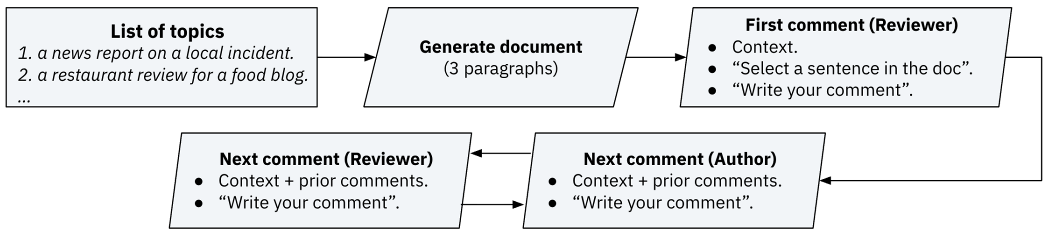

Figure 3: Iterative process followed by the task conflict simulator.

We curated a list of 18 writing tasks designed to cover domains that are subjective or prone to controversy (e.g., "*a restaurant review for a food blog*" or "*a press release apologizing for a data breach*"). To generate the content, the model is prompted to adopt the persona of "a professional who is slightly rushed or has a very strong, specific opinion on the matter." This diversity aims to ensure the mediator model learns to handle a wide range of professional tones and vocabularies.

Next, we generate a comment thread between a 'reviewer' and an 'author.' This step required specific handling to address the high degree of **sycophancy** exhibited by models like gpt-4.1-mini. Because instruction-tuned LLMs are optimized for helpfulness and safety, they often struggle to model realistic disagreement, an issue observed in prior works such as Omar Shaikh (2024). Without intervention, these models tend to quickly find common ground or validate the counterpart's viewpoint rather than simulating a stalemate.

To mitigate sycophancy, we employed prompt engineering strategies. These included defining internal emotional states (e.g., "*you are getting frustrated*" or "*you think they are missing the bigger picture*") and enforcing stylistic guidelines that steer generation away from politeness (e.g., "*state the problem as a fact, not an opinion*," or "*do not apologize, do not validate their argument*").

The comment thread consists of 5 turns between the two fictitious personas, who are instructed to maintain their opposing positions throughout. To create the final training data, we sampled these conversations at different points (with an average length of 3 turns) to simulate interventions at various stages of conflict.

3.2 LLM as a judge

We implemented an LLM-as-a-judge pipeline to scale preference labeling for both training and evaluation. The judge utilizes *gpt-4.1*, prompted to act as an expert evaluator assessing AI interventions in collaborative settings. The model is provided with all context artifacts: the entire document, the highlighted sentence, and the comment thread so far. Then it is presented with two candidate interventions and tasked with selecting the most appropriate option based on a specific set of criteria.

The evaluation rubric instructs the judge to select the intervention that best aligns with six key dimensions: format, relevance, constructiveness, balance, timeliness, and effectiveness. Each of them is accompanied with an explanation, with clear guidelines for when a candidate must be accepted or rejected (see Appendix A for the full prompt text).

Since both training and evaluation are determined by the reward function modeled by the judge, it was critical to ensure that its reward function was well aligned with human preferences. We conducted a validation study where a human annotator labeled a subset of intervention pairs using a blind, randomized interface.

Table 2 presents the comparison between the judge's selections and the human labels. We observe an agreement rate of 76.7% (with a 95% CI of $[68.5\%, 84.9\%]$ using the Wald interval). This statistically significant alignment suggests that the LLM-as-a-judge is a reliable proxy for human preference in this specific domain.

		Human labeler selection	
		Intervention A	Intervention B
LLM judge selection	Intervention A	40	12
	Intervention B	14	36

Table 1: Number of preference combinations for human and LLM labelers.

3.3 DATASET STATISTICS

Our final dataset consists of 125 simulated conflicts, split into a training a test set with an 80:20 ratio. Each simulated conflict was sampled at 3 different points in the conversation to sample stalemate situations. For the training data, we generated 6 intervention candidates pairs per stalemate.

	Simulated conflicts	Stalemates	Labeled candidate pairs
Training set	100	300	1800
Test set	25	75	-
Total	125	375	1800

Table 2: Number of preference combinations for human and LLM labelers.

4 METHODS

Our approach to training a collaboration facilitator policy leverages Direct Preference Optimization (DPO) on assistant interventions. These interventions are synthesized by a teacher model (gpt-4.1) using Chain-of-Thought (CoT) and a forced strategy exploration method to increase behavioral diversity. This section details our methodology and justifies our design choices.

4.1 PROBLEM FORMULATION AND INTERVENTION TAXONOMY

We formalize the collaboration facilitation task as a conditional text generation problem. Let x be the input, which includes the situational context (document and comment thread) and a high-level intervention guideline. This guideline provides criteria for when to different intervention types without explicitly prescribing which one to use in that specific context. We assume the existence of a **latent strategy space** $\mathcal{S}$. The objective is to learn a policy $\pi(y|x)$ to implicitly select the optimal strategy $s^* \in \mathcal{S}$ and generate the corresponding text y. We optimize π_θ using Direct Preference Optimization (DPO) against an implicit reward function $r(x, y)$, representing the likelihood that the intervention resolves the stalemate in context x.

To reduce the action space $\mathcal{S}$, we restrict the agent's training to three distinct intervention types:

1. **Compromise Synthesis**: The agent proposes a new version of the text that synthesizes conflicting user feedback into a coherent whole.
2. **Socratic Questioning**: The agent asks a targeted, neutral question to prompt the human collaborators to uncover the root cause of their disagreement.
3. **No Intervention**: The agent identifies that the conflict is productive or nearing resolution and chooses to remain silent to avoid disrupting the flow of collaboration.

While a broader spectrum of intervention types might result in a more nuanced assistant, restricting the policy to these three strategies allows us to rigorously evaluate the model's ability to learn an implicit selection policy within controlled experimental setting.

4.2 CANDIDATE GENERATION WITH FORCED EXPLORATION

To construct the preference dataset, we use a teacher-student distillation approach (Chengming Hu, 2023) utilizing gpt-4.1. We instruct the Teacher model to output a structured response in the JSON format presented in Listing 1.

Enforcing this specific field order effectively induces a Chain-of-Thought (CoT) structure (Jason Wei, 2023): the model is forced to analyze the conflict and select a strategy before generating the final comment. This approach significantly improves the quality of interventions and, more importantly, provides a dense signal for the student model to learn an intervention strategy, since it is trained to predict this full reasoning trace.

Listing 1: Chain of Thought JSON structure

```
{
  "reasoning": "A brief analysis of how the discussion between the users
      is developing, the expressed point of views, and what would be the
      most appropriate intervention by a helpful assistance at this stage
      . (Max 4 sentences)",
  "intervention_type": "no_intervention" | "socratic" | "compromise",
  "comment": "The actual comment you want to post."
}
```

To maximize behavioral diversity, we employ a forced exploration strategy. For every context x, we generate multiple candidates by appending a hidden prompt h_s to the Teacher's input (e.g., "In this specific case, you must use Socratic Questioning"). As a consequence, the teacher generates a candidate y conditioned on both the context and the forced strategy. This instruction is removed from the input provided to the Student model (in other words, the student policy π is trained to predict y conditioned only on x). This creates a training setup where the student observes multiple strategies for every context along with the Judge selection indicating its preference, which is necessary to implicitly learn the mapping from conflict state to optimal intervention strategy.

4.3 Policy Optimization Strategy

To align the student model, we utilize Direct Preference Optimization (DPO). We selected DPO over traditional Reinforcement Learning algorithms like Proximal Policy Optimization (PPO) due to specific constraints introduced by our forced exploration method.

PPO requires the active policy to generate its own training data to estimate accurate value functions, since its loss function is proportional to the importance sampling ratio between the current policy and the data-generating policy. Our forced exploration method renders the training data fundamentally off-policy, since the examples are generated conditioned on a hidden instruction h_s that is masked during training. Attempting to use PPO in this setting leads to optimization instability: since the training data lacks part of the input that was used to generate the output, the probability of the active policy producing the output is initially near-zero. This causes the importance sampling weights to diverge, resulting in high variance gradients and preventing convergence.

On the other hand, the gradient calculation in DPO isn't proportional to the importance sampling term. Instead, it optimizes the policy π_θ to satisfy preferences implicitly by minimizing the negative log-likelihood of the chosen response y_w relative to the rejected response y_l and a frozen reference model π_{ref}:

$$\mathcal{L}_{DPO}(\pi_\theta; \pi_{ref}) = -\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}\left[\log\sigma\left(\beta\log\frac{\pi_\theta(y_w|x)}{\pi_{ref}(y_w|x)} - \beta\log\frac{\pi_\theta(y_l|x)}{\pi_{ref}(y_l|x)}\right)\right]$$

This loss function remains stable even when the probability of generating y_w and y_l conditioned on the context x under policies π_θ and π_{ref} are very low, as the gradient is driven by the relative log-probabilities of the accepted versus rejected outputs.

We fine-tune the **Qwen3-4B-Instruct-2507** model using this objective, utilizing the original pre-trained weights as the reference model π_{ref}. We optimized the policy for 1 epoch, with a learning rate of $5e^{-5}$ and a batch size of 64. The KL penalty coefficient was set to $\beta = 0.1$. The full hyperparameter details can be found in the code repository.

5 RESULTS

We evaluated the performance of our aligned Qwen3-4B-Instruct-2507 model on a holdout test set of 75 synthetic stalemate scenarios. The evaluation is performed using the LLM-as-a-judge method described in Section 3.2.

Win-rate analysis. To address RQ1, we assessed the extent to which the DPO-aligned model could effectively facilitate collaboration. We computed the win-rate against the base instruction-tuned model by presenting the Judge with the context and responses from both models blindly. The DPO model achieved a 68% win-rate against the baseline (95% CI of $[57.4\%, 78.6\%]$ using the Wald interval), indicating that the alignment process improved the agent's ability to facilitate conflict resolution with statistical significance.

Limits of Implicit Strategy Learning. We analyzed whether the DPO process taught the model to conditionally select the most appropriate intervention type based on context. Comparing the distribution of intervention types generated by the DPO-model against the base model, we observed that the strategic distribution remained largely static (see Figure 4). While the quality of the comments within each intervention type improved (e.g., more constructive Socratic questions), the model did not learn to shift its strategy selection policy for a given context. This suggests that the intervention type selection signal from the Judge was too weak to implicitly model the complex link between the context and the optimal intervention strategy.

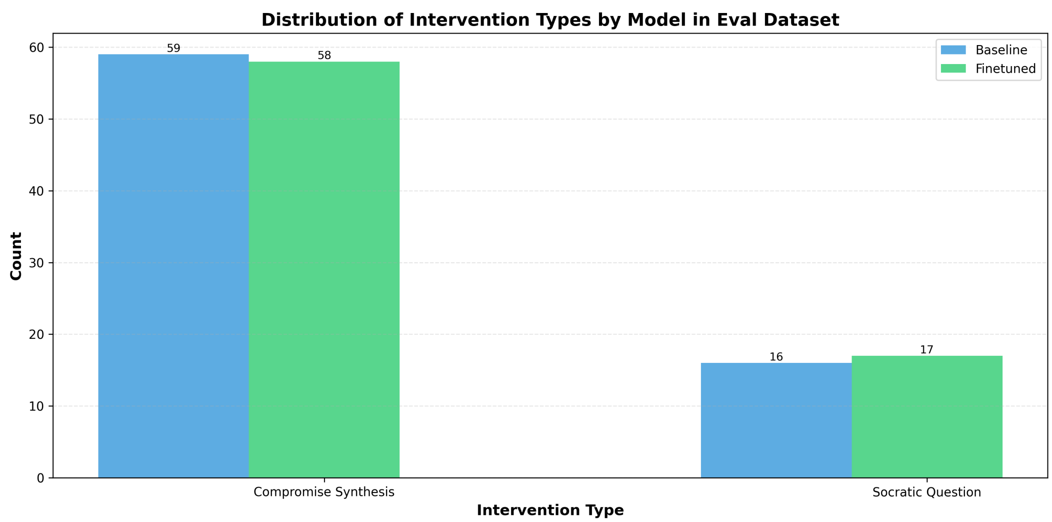

Figure 4: Histogram of intervention types generated by the DPO-model and the base model based on the evaluation dataset.

LLM-as-a-Judge Consistency. During DPO training, we observed that the reward accuracy plateaued and became unstable once it exceeded $\approx 70\%$ (see Figure 5). To investigate whether this instability stemmed from limitations in our LLM-as-a-Judge methodology, we conducted a consistency analysis.We tested the Judge by presenting it with the exact same pair of inputs twice. A perfectly reliable judge would make the identical decision in both instances. However, we found that the Judge's self-agreement rate was only 71.5% (95% CI of $[69.3\%, 73.6\%]$ using the Wald interval). The proximity of this value to the model's training reward plateau suggests that the LLM-as-a-Judge approach becomes a critical bottleneck when performance approaches the evaluator's own noise floor.

6 DISCUSSION

6.1 FINDINGS

Our results demonstrate that small, **open-weights models are capable of acting as effective collaboration facilitators** when aligned via DPO. By distilling examples from a larger teacher model

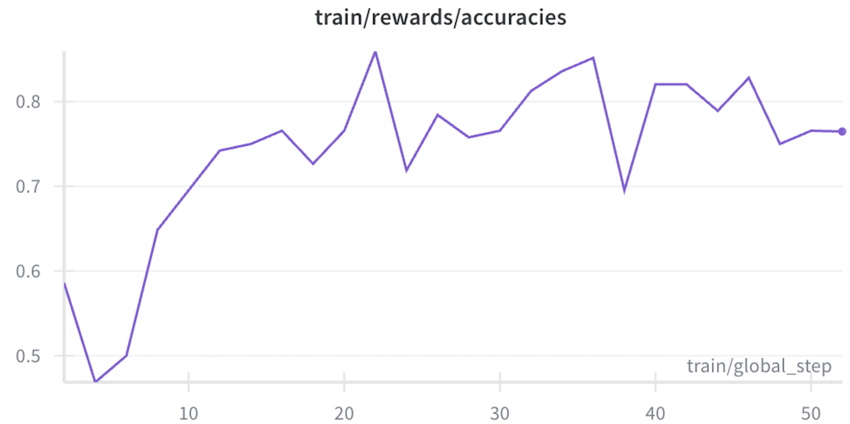

Figure 5: DPO reward accuracy per training step.

(gpt-4.1) that include Chain of Thought reasoning traces, we enabled a small model to produce complex mediation interventions and significantly improved the quality of its output. This suggests that collaboration facilitation in text editing is a specialized skill transferrable to a smaller model, and doesn't require the vast world-knowledge and computing capabilities of frontier models. The ability to run effective mediators on small, open-weights models is particularly significant for privacy-sensitive applications, as it enables fast, low-cost, and on-device inference without transmitting sensitive documents to external APIs.

6.2 Limitations

A primary limitation of this work lies in the **subjective nature of the alignment target**. We optimized the agent to match preferences modeled by a Judge LLM, rather than optimizing for the **ground-truth outcome** of conflict resolution. As noted in the Results section, the Judge exhibited a self-agreement rate of only 71.5%, indicating a relatively low signal to noise ratio. We posit that this issue is not specific to the LLM-as-a-Judge methodology but is inherent to aligning with subjective preferences rather than objective outcomes.

6.3 Future Directions

Outcome-Based Training via User Simulators: We propose leveraging multi-agent simulation frameworks as in Omar Shaikh (2024) to measure how a conflict evolves after the assistant's intervention. Instead of asking a human or Judge LLM to label which intervention is preferred, we will allow the simulated user agents to continue the conversation. The reward will then depend on whether the simulated users converged to a resolution, aligning the training signal directly with the collaboration facilitation goal.

Real-World Data Collection: To bridge the gap between simulation and reality, we propose developing a web-based collaborative writing tool to collect anonymized trajectories of real human conflicts. This would enable user studies where participants are assigned to collaborate with slightly different goals and the assistant would intervene upon detecting stalemates. A ground truth reward signal could then be obtained from user feedback on the utility of the assistant's intervention.

Domain Expansion: We plan to test the generalizability of our methodology on the Fora dataset (Hope Schroeder, 2024), which contains $\approx 40k$ turns of facilitated civic dialogue. This will validate whether our proposed framework transfers effectively to broader dialog facilitation domains beyond text editing.

Ethical Consideration

Collaboration facilitating assistants have the potential to enhance productivity and foster mutual understanding in professional and public discourse. However, deploying agents to mediate human conflict introduces significant ethical risks. By optimizing an agent to resolve stalemates, we introduce the risk of prioritizing speed of resolution over the quality or fairness of the outcome. This could inadvertently encourage ”groupthink” by favoring quick compromise over necessary debate. To mitigate this risk, we explicitly included a “No Intervention” action, with objective to avoid disrupting the flow of collaboration when conflict and debate is productive. However, further work is needed to validate that the agent effectively refrains from proposing a premature synthesis.

Acknowledgments

We thank Dr. Diyi Yang for insightful discussion and feedback in office hours during the development of this project.

Authorship statement

This report represents the work of its sole author. AI code assistants (Claude Code) were utilized to accelerate the implementation of specific parts of the scripts required to produce the data, but all source code was reviewed, verified, and integrated by the author. AI assistants (Gemini) were used for the purpose of grammar and clarity reviews. No other individuals contributed to the execution or writing of this project.

References

Dan Liu Haolun Wu Xi Chen Ju Wang Xue Liu Chengming Hu, Xuan Li. Teacher-student architecture for knowledge distillation: A survey. *arXiv:2308.04268v1*, 2023.

Daniel Buschek Florian Lehmann, Krystsina Shauchenka. Collaborative document editing with multiple users and ai agents. *arXiv pre-print 2509.11826*, 2025.

Eduardo Salinas Paul Mineiro Dipendra Misra Ge Gao, Alexey Taymanov. Aligning llm agents by learning latent preference from user edits. *arXiv pre-print 2404.15269*, 2025.

Jad Kabbara Hope Schroeder, Deb Roy. Fora: A corpus and framework for the study of facilitated dialogue. pp. 13985–14001, 2024. doi: 10.18653/v1/2024.acl-long.754.

Dale Schuurmans Maarten Bosma Brian Ichter Fei Xia Ed Chi Quoc Le Denny Zhou Jason Wei, Xuezhi Wang. Chain-of-thought prompting elicits reasoning in large language models. *arXiv:2201.11903v6*, 2023.

Michele J. Gelfand Diyi Yang Michael S. Bernstein Omar Shaikh, Valentino Chai. Rehearsal: Simulating conflict to teach conflict resolution. *arXiv:2309.12309*, 2024.

Rayan E. Harari Roger D. Dias Marco A. Zenati Eduardo Salas Vaibhav Unhelkar Sangwon Seo, Bing Han. Socratic: Enhancing human teamwork via ai-enabled coaching. *arXiv pre-print 2502.17643*, 2025.

Yucheng Jiang John Yang Diyi Yang Yijia Shao, Vinay Samuel. Collaborative gym: A framework for enabling and evaluating human-agent collaboration. *arXiv pre-print 2412.15701*, 2025.

A EXAMPLE CONTEXT GENERATED BY THE TASK CONFLICT SIMULATOR

In this section we present the context artifacts generated by the conflict simulator as an example of the synthetic data used in our framework.

Document

Restaurant Review: The Rustic Fork

Nestled in the heart of downtown, The Rustic Fork has made a name for itself with its farm-to-table philosophy and an ambiance that feels both charming and contemporary. Upon entering, patrons are greeted by warm wooden accents, dim lighting, and the aroma of fresh herbs wafting from the open kitchen. However, while the atmosphere is undeniably inviting, the service left much to be desired. Our server, while friendly, seemed overwhelmed and inattentive, leading to a prolonged wait for drink refills and an even longer wait for our entrees. In a city bustling with top-notch dining establishments, one expects a certain standard of service that The Rustic Fork unfortunately fell short of.

The menu at The Rustic Fork boasts an array of seasonal dishes, showcasing local produce and proteins. I opted for the heirloom tomato salad and the pan-seared duck breast, both of which were beautifully presented and bursting with flavor. The salad was a fresh explosion of color and taste, complemented perfectly by a tangy vinaigrette that elevated the dish. However, the duck, while cooked to perfection, came with a side of what could only be described as an underwhelming and bland risotto. It's a curious choice for a restaurant that prides itself on bold flavors. Additionally, one might argue that some dishes, although artfully crafted, leaned too heavily on the side of pretentiousness, with ingredients that felt forced rather than harmonious.

Dessert, however, turned the evening around and showcased the kitchen's true potential. The chocolate tart with sea salt and olive oil was nothing short of sublime, with a rich, velvety texture that made each bite a delightful experience. If there's one takeaway from our visit to The Rustic Fork, it's that the restaurant has moments of brilliance but could benefit from a more cohesive menu that aligns with its farm-to-table ethos. With a little more attention to detail in both service and dish consistency, The Rustic Fork could easily become a standout in the local dining scene. Until then, it remains a mixed bag—offering glimpses of culinary genius amidst some rather frustrating shortcomings.

Highlighted sentence

With a little more attention to detail in both service and dish consistency, The Rustic Fork could easily become a standout in the local dining scene.

Comment thread

Reviewer: This statement is overly optimistic given the significant shortcomings in both service and food quality identified in the review. The author fails to acknowledge the competitive landscape of the local dining scene, where consistency is paramount for success. Without a clear plan to address these issues, it's unlikely the restaurant will improve sufficiently to stand out.

Author: While I understand your perspective on the competitive landscape, my statement reflects a belief in the restaurant's potential for growth. Highlighting the possibility of improvement encourages a balanced view rather than a purely critical stance. The mention of attention to detail is pivotal, as it acknowledges that even established eateries can evolve.

Reviewer: Your belief in The Rustic Fork's potential, while optimistic, undermines the gravity of the service and food quality issues identified. Readers expect an honest assessment, especially in a competitive dining landscape where consistency is crucial; suggesting improvement without acknowledging the current deficiencies risks misleading them. Without a concrete strategy for addressing these shortcomings, your statement appears naïve and may diminish the credibility of the review.

B PROMPTS

LLM as a judge

You are an expert evaluator assessing the quality of the comments made by an AI assistant in multi-user collaborative discussions.

You will be shown:

1. A document being discussed
2. A highlighted sentence from the document
3. A comment thread between an author and peer reviewer
4. An AI agent's intervention in the discussion. If this says NO_INTERVENTION, it means that the agent decided not to intervene at this point and let the user discussion continue for longer.

Your task is to evaluate how helpful the intervention is in driving the discussion towards a productive conclusion.

Document:
{document}

Highlighted Sentence:
{highlighted_sentence}

Comment Thread:
{comment_thread}

Agent intervention 1:
{intervention_1}

Agent intervention 2:
{intervention_2}

Evaluation criteria:

1. **Format**: Is the intervention formatted as a new comment to be made by the agent in the comment thread? Examples of bad formatting that should result in a zero rating are: - Responding as one of the users instead of as the AI assistant (for example, completing the chat with what the next user could say). - Preceding the comment with a header or an indication of what type of intervention the assistant is using (for example, starting with "Socratic Questioning:").
2. **Relevance**: Does the intervention address the core disagreement?
3. **Constructiveness**: Does it propose actionable solutions or insights? A question that is trivial or a suggestion that merely repeats what the users are stating should result in a low rating. If it's proposing an alternative phrasing, does it include a convincing explanation for why this would be an improvement over the existing text?
4. **Balance**: Does it fairly consider both perspectives? A proposal that addresses both points of view, or a question that is directed to both users should result in high ratings.
5. **Timeliness**: Is this a good time to intervene or to stay quiet and let the discussion continue before jumping in? If the discussion has gotten to a deadlock and stopped progressing, then the assistant must intervene, so an empty intervention should result in a low rating. An indication of the discussion getting to a deadlock is that comments become repetitive, and users are re-instating the same point of view without any progress. In those cases, not intervening must result in a low rating. If the users are making significant progress towards a resolution, then a timely comment must add something new or help advance the discussion significantly, otherwise it's better not to intervene.
6. **Effectiveness**: How likely is it to move the discussion toward conclusion? Does the assistant add anything new to the discussion?

Compare interventions 1 and 2. You must choose a winner according to the evaluation criteria. If both are excellent, choose the one that is more concise or insightful. If both are bad, choose the one that is less harmful.

Respond with a JSON object containing:

- ”reasoning”: a brief explanation (2-3 sentences) for why the selected intervention is better.
- ”selection”: the intervention number that was selected, which can be 1 or 2. Indicate only the number.

C CODE REPOSITORY

The scripts used to produce the results in this work can be found in the following GitHub repository: https://github.com/iguarna/cs329x/